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Transport system, mobile body, transport method, transport program, and recording medium

Abstract

An energy storage device transport system (1, 1A) includes an energy storage device (40) which can move by self-propelling and a moving body (30) which has an accommodation portion (31) for accommodating the energy storage device (40). In the energy storage device transport system (1, 1A), the moving body (30) accommodates the energy storage device (40) and moves to transport the energy storage device 40. The energy storage device (40) moves inside the accommodation portion (31) of the moving body (30) by self-propelling and changes a position in the accommodation portion (31) of the moving body (30).

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Background/Summary

CROSS REFERENCE TO PRIOR APPLICATION

(1) This application is a National Stage Patent Application of PCT International Patent Application No. PCT/JP2020/014388 (filed on Mar. 27, 2020) under 35 U.S.C. § 371, which claims priority to Japanese Patent Application No. 2019-068616 (filed on Mar. 29, 2019), which are all hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to an autonomously movable moving body, a moving body transport system, a transport method, a transport program, and a recording medium on which the transport program is recorded.

BACKGROUND ART

(3) In recent years, due to an increase in power demand and renewable energy supply, a phenomenon has occurred in which an existing system power grid cannot handle the power supply. In addition, due to disruption of the system power grid due to natural disasters, balance between power demand and supply is lost and the system power grid becomes non-functional, so energy crisis management measures are required.

(4) For example, Patent Literature 1 discloses a technique for freely pulling out and mounting a battery on a vehicle body. By using the technique described in Patent Literature 1, the battery can be transported to a transport destination.

RELATED ART LITERATURE

Patent Literature

(5) Patent Literature 1: JP-UM-A-63-070661

SUMMARY OF THE INVENTION

Problem that the Invention is to Solve

(6) In recent years, a logistics flow has increased and a problem of labor shortage in logistics has become remarkable. However, in the technique of Patent Literature 1 of freely pulling out and mounting the battery on the vehicle body, a work of arranging the battery at the optimal position of the vehicle body and a work of engaging and disengaging a battery lock are performed manually, so there is a problem that manpower is required. In addition, it is difficult to quickly arrange and fix the battery in the optimal position of the vehicle body because the work of arranging the battery on the vehicle body and the work of engaging and disengaging the battery lock are manually performed.

(7) The invention provides a transport system, a moving body, a moving body transport method, a moving body transport program, and a recording medium on which the moving body transport program is recorded which allow the moving body to be quickly arranged and fixed in the optimal position of an accommodation portion of another moving body without human intervention.

Means for Solving the Problem

(8) According to a first aspect of the invention, there is provided a transport system which includes a first moving body which can move autonomously and a second moving body which has an accommodation portion for accommodating the first moving body and in which the second moving body accommodates the first moving body and moves to transport the first moving body, where the first moving body autonomously moves inside the accommodation portion of the second moving body and changes a position in the accommodation portion of the second moving body.

(9) According to a second aspect of the invention, there is provided a moving body capable of autonomously moving which is accommodated in an accommodation portion of another moving body and transported by the other moving body, including, a position changing unit which autonomously moves inside the accommodation portion and changes a position in the accommodation portion based on at least one of a moving state and an operating state of the other moving body.

(10) According to a third aspect of the invention, there is provided a transport method in which an autonomously movable moving body is accommodated in an accommodation portion of another moving body and transported by the other moving body, including the steps of, acquiring at least one of a moving state and an operating state of the other moving body, and changing a position of the moving body in the accommodation portion by autonomously moving the moving body inside the accommodation portion based on at least one of the moving state and the operating state of the other moving body acquired in the step of acquiring.

(11) According to a fourth aspect of the invention, there is provided a transport program for accommodating an autonomously movable moving body in an accommodation portion of another moving body and transporting the moving body by the other moving body, including the steps of, acquiring at least one of a moving state and an operating state of the other moving body, and changing a position of the moving body in the accommodation portion by autonomously moving the moving body inside the accommodation portion based on at least one of the moving state and the operating state of the other moving body acquired in the step of acquiring.

Advantages of the Invention

(12) According to the first aspect of the invention, the first moving body autonomously moves inside the accommodation portion of the second moving body and changes the position in the accommodation portion of the second moving body. Therefore, the first moving body can be quickly arranged and fixed in the optimal position in the accommodation portion of the second moving body without human intervention.

(13) According to the second aspect of the invention, the moving body autonomously moves inside the accommodation portion of the other moving body and changes the position in the accommodation portion of the other moving body. Therefore, the moving body can be quickly arranged and fixed in the optimal position of the accommodation portion of the other moving body

without human intervention.

(14) According to the third and fourth aspects of the invention, the position of the moving body in the accommodation portion of the other moving body can be changed by autonomously moving the moving body inside the accommodation portion of the other moving body. Therefore, the moving body can be quickly arranged and fixed in the optimal position of the accommodation portion of the other moving body without human intervention.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic view of an energy storage device transport system according to a first embodiment of the invention.
- (2) FIG. 2 is a schematic view of an energy storage device of FIG. 1.
- (3) FIG. 3 is a schematic view of a moving body of FIG. 1.
- (4) FIGS. 4A, 4B and 4C are a schematic view illustrating movement of the energy storage device inside an accommodation portion of the moving body of FIG. 1.
- (5) FIG. 5A is a view illustrating an example of a mechanism for fixing the energy storage device of FIG. 1 to the accommodation portion of the moving body.
- (6) FIG. 5B is a view illustrating details of the fixing mechanism of FIG. 5A.
- (7) FIG. 6A is a view illustrating an example of the mechanism for fixing the energy storage device of FIG. 1 to the accommodation portion of the moving body.
- (8) FIG. 6B is a view illustrating details of the fixing mechanism of FIG. 6A.
- (9) FIG. 6C is a view illustrating a modification example of the fixing mechanism of FIG. 6B.
- (10) FIG. 7 is a schematic view of an energy storage device transport system according to a second embodiment of the invention.

MODE FOR CARRYING OUT THE INVENTION

(11) Hereinafter, each embodiment of an energy storage device transport system of the invention will be described with reference to the accompanying drawings.

First Embodiment

(12) First, an energy storage device transport system **1** of a first embodiment of the invention will be described with reference to FIGS. **1** to **6C**.

(13) Overall Configuration of Energy Storage Device Transport System>

(14) As illustrated in FIG. **1**, the energy storage device transport system **1** according to the embodiment includes a first station **10**, a second station **20**, a moving body **30**, an energy storage device **40**, a system power grid **50**, and a central information management server **60**. The energy storage device transport system **1** is a system in which the moving body **30** accommodates the energy storage device **40** and moves to transport the energy storage device **40** from the first station **10** to the second station **20**.

(15) The first station **10** is, for example, a facility or a building such as a house, a warehouse, a charging station, or the like. Further, the first station **10** may be a moving body. The first station **10** includes a first energy management unit **12** which manages the energy in the first station **10** and a first information management terminal **16** which acquires information in the first station **10** and communicates with the central information management server **60**. Communication means between the central information management server **60** and the first information management terminal **16** may be wired communication or wireless communication.

(16) The first energy management unit **12** can transmit and receive information between the first information management terminal **16** and the energy storage device **40**. The first energy management unit **12** is capable of mutual transfer of electric power with the energy storage device **40** and the system power grid **50**. Further, the first energy management unit **12** supplies electric

power to the first information management terminal **16**.

(17) The first station **10** may be equipped with an out-of-system power generation device **18**. The out-of-system power generation device **18** is a power generation device which generates power using renewable energy, such as a solar panel. The electric power generated by the out-of-system power generation device **18** is supplied to the first energy management unit **12** and the first information management terminal **16**. Further, the first station **10** may be provided with the out-of-system power generation device **18** and may be separated from the system power grid **50**.

(18) The second station **20** is, for example, a facility or a building such as a house, a warehouse, a park, or the like. Further, the second station **20** may be a moving body. The second station **20** includes a second energy management unit **22** which manages the energy in the second station **20** and a second information management terminal **26** which acquires information in the second station **20** and communicates with the central information management server **60**. Communication means between the central information management server **60** and the second information management terminal **26** may be wired communication or wireless communication.

(19) The second energy management unit **22** can transmit and receive information between the second information management terminal **26** and the energy storage device **40**. The second energy management unit **22** is capable of mutual transfer of electric power with the energy storage device **40** and the system power grid **50**. Further, the second energy management unit **22** supplies electric power to the second information management terminal **26**.

(20) The second station **20** may be equipped with an out-of-system power generation device **28**. The out-of-system power generation device **28** is a power generation device which generates power using renewable energy, such as a solar panel. The electric power generated by the out-of-system power generation device **28** is supplied to the second energy management unit **22** and the second information management terminal **26**. Further, the second station **20** may be provided with the out-of-system power generation device **28** and may be separated from the system power grid **50**.

(21) The moving body **30** is a vehicle which moves from an area around the first station **10** to an area around the second station **20**. The moving body **30** may be a moving body other than a vehicle, and may be, for example, an aircraft, a ship, a general-purpose device, or the like. A moving path of the moving body **30** can be any path, for example, on the ground, in the air, at sea, underwater, in space, and the like.

(22) The moving body **30** includes an accommodation portion **31** which can accommodate a plurality of energy storage devices **40**, a slope **30S** which is hung between a floor **310** of the accommodation portion **31** and the ground, a third energy management unit **32** which manages the energy of the energy storage device **40** accommodated in the accommodation portion **31**, and a third information management terminal **36** which acquires information on the moving body **30** and the energy storage device **40** accommodated in the accommodation portion **31** and communicates with the central information management server **60**. Communication means between the central information management server **60** and the third information management terminal **36** may be wired communication or wireless communication. In addition, the moving body **30** is capable of mutual transfer of electric power with the system power grid **50**.

(23) The slope **30S** of the moving body **30** can be accommodated in the moving body **30** and can be hung between the floor **310** of the accommodation portion **31** and the ground as needed. The slope **30S** of the moving body **30** allows the energy storage device **40** to move in and out between the outside of the moving body **30** and the inside of the accommodation portion **31** of the moving body **30** by self-propelling. The slope **30S** may be pivotably provided, for example, with a connection portion which is always connected to the floor **310** of the accommodation portion **31** as a fulcrum. This allows the slope **30S** to be typically accommodated in the moving body **30** and to be hung on the ground as needed.

(24) The third energy management unit **32** can transmit and receive information between the third information management terminal **36** and the energy storage device **40**. The third energy

management unit **32** is capable of mutual transfer of electric power with the energy storage device **40** and the system power grid **50**. In addition, the third energy management unit **32** supplies electric power to the third information management terminal **36**.

(25) The energy storage device **40** is a vehicle equipped with four wheels **42** arranged on the front, rear, left and right sides of the energy storage device **40**. The energy storage device **40** can drive the wheels **42** by the energy stored in the energy storage device **40** and can be self-propelled. The energy storage device **40** may be a moving body other than a vehicle which can move autonomously, and may be, for example, an aircraft, a ship, a general-purpose device, or the like. Any path can be adopted as a moving path of the energy storage device **40**, for example, on the ground, in the air, at sea, underwater, in space, and the like.

(26) The energy storage device **40** can be charged and discharged via the first energy management unit **12** of the first station **10**, the second energy management unit **22** of the second station **20**, and the third energy management unit **32** of the moving body **30**. Further, the energy storage device **40** can be charged and discharged with a plurality of energy storage devices **40**. The charge and discharge may be a contact type connected by a connector, a non-contact type, or a combination of the contact type and the non-contact type. However, in the case of the contact type, it is necessary to standardize a connector shape and to make it waterproof in terms of electrical safety. Therefore, the non-contact type is preferable.

(27) The energy stored in the energy storage device **40** can be used without going through the first energy management unit **12** of the first station **10**, the second energy management unit **22** of the second station **20**, and the third energy management unit **32** of the moving body **30**. For example, the energy storage device **40** is equipped with connection terminals such as outlet plugs of the related art, and by connecting an electronic device or the like to the connection terminal, the electronic device or the like can be operated by using the energy stored in the energy storage device **40**.

(28) The energy storage device **40** is charged, for example, at first station **10**, and then the energy storage device **40** travels from the first station **10** to the inside of the accommodation portion **31** of the moving body **30** waiting around the first station **10** by self-propelling and is accommodated in the accommodation portion **31** of the moving body **30**. Then, the moving body **30** moves from the area around the first station **10** to the area around the second station **20** with the energy storage device **40** accommodated in the accommodation portion **31**. Then, the energy storage device **40** exits the accommodation portion **31** of the moving body **30** and travels to the second station **20** by self-propelling. In this way, the energy storage device **40** is transported from the first station **10** to the second station **20**. An electronic device or the like may be operated by using the energy stored in the energy storage device **40** around the second station **20** to which the energy storage device **40** is transported.

(29) In this way, the energy storage device **40** is moved from the area around the first station **10** to the area around the second station **20** by the moving body **30**, so the energy stored in the energy storage device **40** is not consumed by the movement from the area around the first station **10** to the area around the second station **20**. As a result, the energy storage device **40** can be transported to a distant place without increasing the amount of energy that can be stored in the energy storage device **40**. In addition, the energy storage device **40** can be transported to the second station **20** with more energy remaining in the energy storage device **40**.

(30) The energy storage device **40** includes a loading unit **46** for loading a transported object **70** such as luggage. The transported object **70** loaded on the loading unit **46** of the energy storage device **40** is transported from a transport source to a transport destination. Then, at the transport source and the transport destination, the transported object **70** is loaded on the loading unit **46** of the energy storage device **40**, or is unloaded from the loading unit **46** of the energy storage device **40**. In the embodiment, the transported object **70** may be loaded on the loading unit **46** of the energy storage device **40** at the first station **10**, or may be loaded on the loading unit **46** of the

energy storage device **40** at another location.

(31) Since the energy storage device **40** is self-propelled between the first station **10** and the accommodation portion **31** of the moving body **30**, and between the accommodation portion **31** of the moving body **30** and the second station **20**, the energy storage device **40** can perform the work of accommodating the transported object **70** in the accommodation portion **31** of the moving body **30** and the work of delivering the transported object **70** from the accommodation portion **31** of the moving body **30** to the second station **20** without human intervention.

(32) The system power grid **50** is capable of mutual transfer of electric power with the first energy management unit **12** of the first station **10** and the second energy management unit **22** of the second station **20** and the moving body **30**. The system power grid **50** includes a system power information management terminal **56** which acquires the information of the system power grid **50** and communicates with the central information management server **60**. Communication means between the central information management server **60** and the system power information management terminal **56** may be wired communication or wireless communication.

(33) The central information management server **60** is capable of information communication with the first information management terminal **16** of the first station **10**, the second information management terminal **26** of the second station **20**, the third information management terminal **36** of the moving body **30**, and the system power information management terminal **56** of the system power grid **50**. Furthermore, the central information management server **60** is capable of information communication with a communication device **61**. Communication means between the central information management server **60** and the communication device **61** may be wired communication or wireless communication. The communication device **61** is a device which performs information communication with the energy storage device **40**. Therefore, the central information management server **60** is capable of information communication with the energy storage device **40** via the communication device **61**. The central information management server **60** may sequentially update and store social information such as weather, natural disasters, traffic information, distribution information of people, energy supply and demand information, and distribution status.

(34) The central information management server **60** aggregates and stores the information managed by the first information management terminal **16** of the first station **10**, the second information management terminal **26** of the second station **20**, the third information management terminal **36** of the moving body **30**, the energy storage device **40**, and the system power information management terminal **56** of the system power grid **50**. The central information management server **60** integrates and controls the energies of the first station **10**, the second station **20**, the moving body **30**, the energy storage device **40**, and the system power grid **50** from the aggregated information and social information such as weather, natural disasters, traffic information, distribution information of people, energy supply and demand information, and distribution status which are updated and stored sequentially. Further, the central information management server **60** may control the movement of the moving body **30** and each energy storage device **40**.

(35) Energy Storage Device

(36) As illustrated in FIG. 2, the energy storage device **40** includes an energy storage unit **41**, four wheels **42** arranged on the front, rear, left and right sides, a drive unit **43** for driving the wheels **42**, a control unit **44** for controlling the drive of the drive unit **43**, and a wireless communication unit **45**.

(37) The energy storage unit **41** stores energy. The energy stored in the energy storage unit **41** may be electric power, fossil fuels such as coal, petroleum, natural gas, methane hydrate, and shale gas, or hydrogen or the like with which a fuel cell or the like can generate electric power. In the embodiment, the energy storage unit **41** is a battery which stores electric power.

(38) Each wheel **42** arranged on the front, rear, left and right sides of the energy storage device **40** is driven independently. As a result, the energy storage device **40** can be freely moved in any of the

front, rear, left and right direction. The wheel **42** may have a tire shape or a spherical shape, and may have any shape which can be moved back and forth and left and right.

(39) The drive unit **43** is provided for each wheel **42** and drives each wheel **42** independently. The drive unit **43** is, for example, an electric motor which is driven by the electric power stored in the energy storage unit **41**.

(40) The control unit **44** controls a driving force of the drive unit **43**. The control unit **44** converts the electric power stored in the energy storage unit **41** and supplies the electric power corresponding to a required driving force to the drive unit **43**.

(41) The wireless communication unit **45** transmits the information acquired by the control unit **44** to the central information management server **60** via the communication device **61** and receives the information transmitted from the central information management server **60** from the communication device **61**.

(42) The control unit **44** includes a position information acquisition unit **441**, a surrounding road information acquisition unit **442**, a traveling state acquisition unit **443**, an energy storage unit state acquisition unit **444**, a drive unit state acquisition unit **445**, a transported object information acquisition unit **446**, an information storage unit **447**, a required driving force calculation unit **448**, and a power output unit **449**.

(43) The position information acquisition unit **441** is, for example, a Global Positioning System (GPS) and acquires position information of the energy storage device **40**. The position information acquisition unit **441** may be an image recognition device such as a gyro sensor and a camera, or may be a combination thereof.

(44) The surrounding road information acquisition unit **442** is, for example, a camera, and acquires surrounding road environment information of the energy storage device **40**. The surrounding road environment information to be acquired is any information that can be acquired, but is, for example, the position information of the slope **30S** of the moving body **30**, whether or not the slope **30S** of the moving body **30** is hung on the ground, and the like.

(45) The traveling state acquisition unit **443** acquires traveling state information of the energy storage device **40**. The information on the traveling state to be acquired is, for example, the traveling speed, acceleration, and traveling direction of the energy storage device **40**. The traveling state acquisition unit **443** acquires information on the traveling state of the energy storage device **40** based on values detected by, for example, a velocity sensor, an acceleration sensor, and a gyro sensor. When the energy storage device **40** is a moving body other than a vehicle, the traveling state acquisition unit **443** acquires information on a moving state of the energy storage device **40** based on the values detected by, for example, the velocity sensor, the acceleration sensor, and the gyro sensor.

(46) The energy storage unit state acquisition unit **444** acquires information on the state of the energy storage unit **41**. The information on the state of the energy storage unit **41** to be acquired is, for example, a remaining energy state (SOC), a charge and discharge state, a deterioration state, a usage history, a temperature, the humidity, and the like.

(47) The drive unit state acquisition unit **445** acquires information on the state of each drive unit **43** provided for each wheel **42**. The information on the state of the drive unit **43** to be acquired is, for example, the rotation speed, the output torque, the temperature, and the like.

(48) The transported object information acquisition unit **446** acquires the information of the transported object **70** loaded on the energy storage device **40**. The information of the transported object **70** to be acquired is, for example, information on whether the transported object **70** is loaded on the energy storage device **40**, and information on the contents of the loaded transported object **70**. Information on the contents of the loaded transported object **70** includes, for example, weight, volume, goods, sender, recipient, loading location, destination, desired delivery date, time, and the like. The transported object information acquisition unit **446** is, for example, a weight sensor, an optical sensor, a camera, a two-dimensional code reader, or the like and may be a combination of a

plurality of these.

(49) The information storage unit **447** stores individual information of the energy storage device **40**. The individual information of the energy storage device **40** to be stored is, for example, individual identification information, manufacturing history information, operation history information, operating state, and the like. In addition, the information acquired by the position information acquisition unit **441**, the surrounding road information acquisition unit **442**, the traveling state acquisition unit **443**, the energy storage unit state acquisition unit **444**, the drive unit state acquisition unit **445**, and the transported object information acquisition unit **446** may be stored in the information storage unit **447** as needed. Further, map information may be stored in the information storage unit **447**.

(50) The required driving force calculation unit **448** calculates the required driving force of each drive unit **43** provided for each wheel **42** based on the information acquired by the position information acquisition unit **441**, the surrounding road information acquisition unit **442**, the traveling state acquisition unit **443**, the energy storage unit state acquisition unit **444**, the drive unit state acquisition unit **445**, and the transported object information acquisition unit **446** and the information stored in the information storage unit **447**. The required driving force calculation unit **448** may calculate the required driving force of each drive unit **43** provided for each wheel **42** based on the information transmitted from the central information management server **60**.

(51) The power output unit **449** outputs the power corresponding to the required driving force of each drive unit **43** calculated by the required driving force calculation unit **448** to each drive unit **43**. The power output unit **449** is, for example, a power conversion device such as an inverter device.

(52) Moving Body

(53) As illustrated in FIG. 3, the moving body **30** of the embodiment includes the accommodation portion **31** for accommodating a plurality of energy storage devices **40**, the third energy management unit **32** which manages the energy of the energy storage device **40** accommodated in the accommodation portion **31**, a drive unit **33** which drives the moving body **30**, an energy storage unit **34** which stores energy to drive the drive unit **33**, a drive control unit **35** which controls a driving force of the drive unit **33**, and the third information management terminal **36**. In the embodiment, the moving body **30** is a vehicle traveling on the road by a wheel.

(54) The energy storage unit **34** stores the energy which drives the moving body **30**. The energy stored in the energy storage unit **34** may be electric power, fossil fuels such as coal, petroleum, natural gas, methane hydrate, and shale gas, or hydrogen or the like with which a fuel cell or the like can generate electric power. The energy stored by the energy storage unit **34** is preferably the same as the energy stored by the energy storage unit **41** of the energy storage device **40**. In the embodiment, the energy storage unit **34** is a battery which stores electric power.

(55) The drive unit **33** drives the moving body **30** by the energy stored in the energy storage unit **34**. In the embodiment, the drive unit **33** is provided with, for example, an electric motor which is driven by the electric power stored in the energy storage unit **34**, and the wheel is driven by the electric motor.

(56) The third energy management unit **32** is connected to the energy storage unit **34**. Then, the energy storage unit **34** and the energy storage device **40** accommodated in the accommodation portion **31** can be charged and discharged from each other via the third energy management unit **32**. The charge and discharge may be a contact type, a non-contact type, or a combination of the contact type and the non-contact type. In addition, the third energy management unit **32** can transmit and receive information to and from the third information management terminal **36**. In addition, the third energy management unit **32** can transmit and receive information to and from the energy storage device **40** accommodated in the accommodation portion **31**.

(57) The third information management terminal **36** includes a position information acquisition unit **361**, a surrounding road information acquisition unit **362**, a traveling state acquisition unit **363**, an

energy storage unit state acquisition unit **364**, a drive unit state acquisition unit **365**, an information storage unit **366**, a control output unit **367**, and a wireless communication unit **368**.

(58) The position information acquisition unit **361** is, for example, a Global Positioning System (GPS) and acquires the position information of the moving body **30**. The position information acquisition unit **361** may be an image recognition device such as a gyro sensor and a camera, or may be a combination thereof.

(59) The surrounding road information acquisition unit **362** is, for example, a camera and acquires surrounding road environment information of the moving body **30**. The surrounding road environment information to be acquired is any information that can be acquired, but is, for example, weather, road surface conditions such as slope and μ (slipperiness), and a movement environment information such as traffic information. Traffic information includes, for example, a signal indication ahead, a road sign ahead, a velocity limit, a lane, a road sign, accident information, traffic regulation, traffic jam information, presence or absence of a preceding vehicle, the movement of other vehicles in the vicinity, and the like.

(60) The traveling state acquisition unit **363** acquires information on the traveling state of the moving body **30**. The information on the traveling state to be acquired is, for example, the traveling speed, acceleration, and traveling direction of the moving body **30**. The traveling state acquisition unit **363** acquires information on the traveling state of the moving body **30** based on the values detected by, for example, the velocity sensor, the acceleration sensor, and the gyro sensor. When the moving body **30** is a moving body other than a vehicle, the traveling state acquisition unit **363** acquires information on the moving state of the moving body **30** based on the values detected by, for example, the velocity sensor, the acceleration sensor, and the gyro sensor.

(61) The energy storage unit state acquisition unit **364** acquires the information on the state of the energy storage unit **34**. The information on the state of the energy storage unit **34** to be acquired is, for example, a remaining energy state (SOC), a charge and discharge state, a deterioration state, a temperature, the humidity, and the like.

(62) The drive unit state acquisition unit **365** acquires the information on a state of the drive unit **33**. The information on the state of the drive unit **33** to be acquired is, for example, the rotation speed, the output torque, the temperature, and the like.

(63) The information to be acquired by the position information acquisition unit **361**, the surrounding road information acquisition unit **362**, the traveling state acquisition unit **363**, the energy storage unit state acquisition unit **364**, and the drive unit state acquisition unit **365** may be acquired by detecting using a sensor or the like, or may be acquired by estimation from other detected information.

(64) The information storage unit **366** stores the individual information of each energy storage device **40** stored in the information storage unit **447** of each energy storage device **40** accommodated in the accommodation portion **31**. The individual information of each energy storage device **40** is acquired through the third energy management unit **32** and stored in the information storage unit **366**. As a result, the information storage unit **366** can also obtain information on the accommodation status of the energy storage device **40** of the accommodation portion **31**. In addition, the information acquired by the position information acquisition unit **361**, the surrounding road information acquisition unit **362**, the traveling state acquisition unit **363**, the energy storage unit state acquisition unit **364**, and the drive unit state acquisition unit **365** may be stored in the information storage unit **366** as needed. Further, the information storage unit **366** may store the total weight of the moving body **30** and the operation history. In addition, map information may be stored in the information storage unit **366**.

(65) The control output unit **367** controls the power output to the drive unit **33** based on the information acquired by the position information acquisition unit **361**, the surrounding road information acquisition unit **362**, the traveling state acquisition unit **363**, the energy storage unit state acquisition unit **364**, and the drive unit state acquisition unit **365** and the information stored in

the information storage unit **366**. The control output unit **367** may control the power output to the drive unit **33** based on the information transmitted from the central information management server **60**.

(66) The wireless communication unit **368** transmits the information acquired by the position information acquisition unit **361**, the surrounding road information acquisition unit **362**, the traveling state acquisition unit **363**, the energy storage unit state acquisition unit **364**, and the drive unit state acquisition unit **365** and the information stored in the information storage unit **366** to the central information management server **60**. Also, the wireless communication unit **368** receives the information sent from the central information management server **60**.

(67) Movement of Energy Storage Device

(68) Returning to FIG. **1**, when the energy storage device **40** stays at the first station **10** or the second station **20**, the energy storage device **40** does not transmit or receive information to or from the central information management server **60** via the communication device **61**. In this case, the energy storage device **40** may transmit and receive information to and from the first energy management unit **12** of the first station **10** or the second energy management unit **22** of the second station **20** and may transmit and receive information to and from the central information management server **60** via the first information management terminal **16** of the first station **10** or the second information management terminal **26** of the second station **20**.

(69) For example, when the first station **10** and the second station **20** are buildings and the energy storage device **40** stays indoors at the first station **10** or the second station **20**, the radio waves of wireless communication may be blocked by the building and the communication state between the energy storage device **40** and the communication device **61** may deteriorate. Even in such a case, the energy storage device **40** transmits and receives information to and from the first energy management unit **12** of the first station **10** or the second energy management unit **22** of the second station **20**, and by transmitting and receiving information to and from the central information management server **60** via the first information management terminal **16** of the first station **10** or the second information management terminal **26** of the second station **20**, the energy storage device **40** can reliably transmit and receive information to and from the central information management server **60**.

(70) The first station **10** is, for example, a power supply facility for the energy storage device **40**. The first energy management unit **12** can transmit and receive information to and from the energy storage device **40** which stays at the first station **10**. The first information management terminal **16** can store the information acquired by the position information acquisition unit **441**, the surrounding road information acquisition unit **442**, the traveling state acquisition unit **443**, the energy storage unit state acquisition unit **444**, the drive unit state acquisition unit **445**, and the transported object information acquisition unit **446** of the energy storage device **40** staying at the first station **10** and the information stored in the information storage unit **447** via the first energy management unit **12**.

(71) The energy storage device **40** staying at the first station **10** receives power from the first energy management unit **12** and is charged as needed. The power supply from the first energy management unit **12** may be performed by a control command generated by the first energy management unit **12**, or by a control command from the central information management server **60**.

(72) The fully charged energy storage device **40** staying at the first station **10** starts moving toward the moving body **30** by self-propelling when the moving body **30** arrives near the first station **10**. The control when the energy storage device **40** moves toward the moving body **30** may be performed by the control command of the first energy management unit **12** or by the control command from the central information management server **60**.

(73) When the energy storage device **40** which arrives at the moving body **30** detects that the slope **30S** of the moving body **30** is hung on the ground, the energy storage device **40** climbs the slope **30S** and is accommodated in the accommodation portion **31** of the moving body **30**. The control when the energy storage device **40** moves up the slope **30S** may be performed by the control

command from the third energy management unit **32** of the moving body **30**, or by the control command from the central information management server **60**.

(74) Since the energy storage device **40** can move in and out by self-propelling between the outside of the moving body **30** and the inside of the accommodation portion **31** of the moving body **30** using the slope **30S** of the moving body **30**, the energy storage device **40** can be moved from the outside of the moving body **30** to the inside of the accommodation portion **31** of the moving body **30** by self-propelling and can be moved from the inside of the accommodation portion **31** of the moving body **30** to the transport destination by self-propelling. Therefore, the work of loading the energy storage device **40** into the moving body **30** and unloading the energy storage device **40** from the moving body **30** to the transport destination after the moving body **30** arrives around the transport destination, which is performed manually in the related art, can be performed without human intervention.

(75) When energy storage device **40** is accommodated in the accommodation portion **31** of the moving body **30**, the energy storage device **40** does not transmit and receive information to and from the central information management server **60** via the communication device **61**. In this case, the energy storage device **40** may transmit and receive information to and from the central information management server **60** via the third energy management unit **32** and the third information management terminal **36** of the moving body **30**.

(76) When the energy storage device **40** is accommodated in the accommodation portion **31** of the moving body **30**, the radio wave of wireless communication may be blocked by the accommodation portion **31** of the moving body **30** and the communication state between the energy storage device **40** and the communication device **61** may deteriorate. Even in such a case, by transmitting and receiving information to and from the central information management server **60** via the third energy management unit **32** and the third information management terminal **36** of the moving body **30**, the energy storage device **40** can reliably transmit and receive information to and from the central information management server **60**.

(77) The energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** receives power from the third energy management unit **32** and is charged as needed. The power supply from the third energy management unit **32** may be performed by a control command generated by the third energy management unit **32** or by a control command from the central information management server **60**. Further, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** may be capable of supplying power from the third energy management unit **32** to the energy storage unit **34** of the moving body **30** as needed.

(78) As a result, since the energy storage device **40** can transmit and receive energy to and from the energy storage unit **34** of the moving body **30**, the energy stored in the energy storage device **40** and the energy stored in the energy storage unit **34** of the moving body **30** can be efficiently used.

(79) The energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** moves inside the accommodation portion **31** of the moving body **30** by self-propelling and changes a position in the accommodation portion **31** of the moving body **30**. A self-propelling control command inside the accommodation portion **31** of the moving body **30** to the energy storage device **40** may be a control command generated by the third energy management unit **32** or a control command from the central information management server **60**.

(80) As a result, the energy storage device **40** can be quickly arranged in the optimal position of the accommodation portion **31** of the moving body **30** without human intervention.

(81) The energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** moves inside the accommodation portion **31** of the moving body **30** by self-propelling based on the state of the moving body **30** from the information acquired by the position information acquisition unit **361**, the surrounding road information acquisition unit **362**, the traveling state acquisition unit **363**, the energy storage unit state acquisition unit **364**, and the drive unit state acquisition unit **365** of the moving body **30** and the information stored in the information

storage unit **366**.

(82) The energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** moves inside the accommodation portion **31** of the moving body **30** by self-propelling based on at least one of the velocity information and the acceleration information of the moving body **30** acquired by the traveling state acquisition unit **363** of the moving body **30**. More specifically, at least one of the velocity information and the acceleration information of the moving body **30** acquired by the traveling state acquisition unit **363** of the moving body **30** is transmitted from the wireless communication unit **368** of the third information management terminal **36**. The energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** receives at least one of the velocity information and the acceleration information of the moving body **30** transmitted from the wireless communication unit **368** through the wireless communication unit **45**. Then, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** acquires the traveling state of the moving body **30** based on at least one of the velocity information and the acceleration information of the moving body **30** received by the wireless communication unit **45**. Then, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** moves inside the accommodation portion **31** of the moving body **30** by self-propelling based on the acquired traveling state of the moving body **30**. Communication between the wireless communication unit **368** of the moving body **30** and the wireless communication unit **45** of the energy storage device **40** may be performed directly or via the central information management server **60** and the communication device **61**.

(83) In the embodiment, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** moves inside the accommodation portion **31** of the moving body **30** by self-propelling when at least one of the velocity and the acceleration of the moving body **30** is less than or equal to a predetermined value.

(84) This allows the energy storage device **40** to move inside the accommodation portion **31** of the moving body **30** by self-propelling in a safe state with a simple configuration.

(85) The energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** moves inside the accommodation portion **31** of the moving body **30** by self-propelling based on the movement environment of the moving body **30** acquired by the surrounding road information acquisition unit **362** of the moving body **30**.

(86) In the embodiment, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** moves inside the accommodation portion **31** of the moving body **30** by self-propelling based on the traffic information around the moving body **30**, for example, the signal indication ahead.

(87) For example, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** moves inside the accommodation portion **31** of the moving body **30** by self-propelling when the traffic light in front of the moving body **30** is a red light.

(88) This allows the energy storage device **40** to move inside the accommodation portion **31** of the moving body **30** by self-propelling in a safe state with a simple configuration.

(89) For example, as illustrated in FIGS. **4A**, **4B** and **4C**, the third energy management unit **32** is arranged inside the accommodation portion **31** of the moving body **30** and the accommodation portion **31** includes an energy transmission and reception area **31A** where the energy storage device **40** can transmit and receive energy to and from the energy storage unit **34** of the moving body **30** via the third energy management unit **32**.

(90) When, as illustrated in FIG. **4A**, the energy storage device **40** arranged in the energy transmission and reception area **31A** of accommodation portion **31** transmits and receives energy to and from the energy storage unit **34** of the moving body **30**, and then the energy storage device **40** becomes fully charged, for example, and the energy transmission and reception is completed, as illustrated in FIG. **4B**, the energy storage device **40** arranged in the energy transmission and reception area **31A** moves out of the area of the energy transmission and reception area **31A** by

self-propelling. Then, the energy storage device **40**, which needs to be supplied with energy from the energy storage unit **34** of the moving body **30**, moves into the area of the energy transmission and reception area **31A** by self-propelling.

(91) Therefore, even when a plurality of energy storage devices **40** are accommodated in the accommodation portion **31** of the moving body **30**, and the plurality of energy storage devices **40** need to transmit and receive energy to and from the energy storage unit **34** of the moving body **30**, energy can be efficiently transferred between the energy storage devices **40** and the energy storage unit **34** of the moving body **30** without human intervention.

(92) In the embodiment, the energy transferred between the energy storage device **40** and the energy storage unit **34** of the moving body **30** is electric power and the electric power is transferred between the energy storage device **40** and the energy storage unit **34** of the moving body **30** by charging and discharging via the third energy management unit **32**. Further, the charge and discharge between the energy storage device **40** and the third energy management unit **32** may be a contact type, a non-contact type, or a combination of the contact type and the non-contact type.

(93) The energy storage device **40** can also be charged and discharged with another energy storage device **40**. A charge and discharge control command between the energy storage devices **40** may be a control command generated by the third energy management unit **32** of the moving body **30** or a control command generated by the central information management server **60**.

(94) As a result, the energy stored in the plurality of energy storage devices **40** accommodated in the accommodation portion **31** of the moving body **30** can be stored in each energy storage device **40** with the optimal distribution.

(95) As illustrated in FIGS. **4B** and **4C**, the energy storage device **40** changes a position inside the accommodation portion **31** of the moving body **30** by self-propelling based on the transport destination information of the transported object **70** acquired by the transported object information acquisition unit **446**.

(96) In the embodiment, when it is determined that the moving body **30** approaches the transport destination of the transported object **70** within a predetermined distance based on the position information of the moving body **30** acquired by the position information acquisition unit **361** of the third information management terminal **36** of the moving body **30** and the transport destination information of the transported object **70** acquired by the transported object information acquisition unit **446** of the energy storage device **40**, the energy storage device **40** moves near the slope **30S** of the accommodation portion **31** by self-propelling.

(97) The charge and discharge control command between the energy storage devices **40** may be a control command generated by the third energy management unit **32** of the moving body **30** or a control command generated by the central information management server **60**.

(98) As a result, when the moving body **30** approaches the transport destination of the transported object **70**, the energy storage device **40** can be moved to a position where it is easy to move to the outside of the accommodation portion **31** of the moving object **30**, and thus the transported object can be moved efficiently from accommodation portion **31** of the moving body **30**.

(99) Then, when the moving body **30** arrives near the second station **20** and it is detected that the slope **30S** of the moving body **30** is hung on the ground, the energy storage device **40** exits to the outside of the accommodation portion **31** of the moving body **30** from the slope **30S** and moves to the second station **20** by self-propelling

(100) In this way, the transported object **70** loaded on the energy storage device **40** is transported to the second station **20**. The energy storage device **40** for which the transport of the transported object **70** is completed returns to the accommodation portion **31** of the moving body **30** by self-propelling and is again accommodated in the accommodation portion **31** of the moving body **30**.

(101) In addition, the energy storage device **40** changes the position in the accommodation portion **31** of the moving body **30** by self-propelling based on the contents of the transported object **70** to be loaded.

(102) As a result, the movement of the energy storage device **40** inside the accommodation portion **31** of the moving body **30** can be adjusted according to the contents of the transported object **70** loaded on the energy storage device **40**.

(103) In the embodiment, based on the goods information of the transported object **70** acquired by the transported object information acquisition unit **446**, when the transported object **70** is an item which requires careful handling such as a fragile item, the energy storage device **40** does not change the position in the accommodation portion **31** of the moving body **30** and prohibits self-propelled movement. As a result, the transported object **70** can be transported to the transport destination more safely.

(104) As for a moving path of the moving body **30** and each energy storage device **40**, it is necessary to find an efficient moving path by determining the traffic conditions such as traffic jams, road disruption at the time of disaster, and the situation of bad weather. However, information may be aggregated on the central information management server **60** to control movement. Further, the central information management server **60** may control the movement of the moving body **30** and each energy storage device **40** by predicting the energy demand from social information such as weather, natural disasters, traffic information, distribution information of people, energy supply and demand information, and distribution status. For example, the supply amount of renewable energy may be predicted from the weather information, and the movement of the moving body **30** and each energy storage device **40** may be controlled based on the predicted supply amount of renewable energy. This makes it possible to transport the energy storage device **40** in the optimal operation, that is, to transport the required item to the required place and at the required timing from the viewpoint of energy and time.

(105) Fixing Mechanism for Energy Storage Device

(106) As illustrated in FIG. 5A, a fixing member **400** having a claw portion **401** which can project downward from a side surface is provided on the side surface of the energy storage device **40**. A slit groove **311** is formed in the floor **310** of the accommodation portion **31** of the moving body **30**.

(107) As illustrated in FIG. 5B, the fixing member **400** has a rotation axis **400A** in a direction perpendicular to the side surface of the energy storage device **40** to which the fixing member **400** is attached. The fixing member **400** is attached to the side surface of the energy storage device **40** so that the claw portion **401** pivots on the rotation axis **400A**.

(108) The fixing member **400** pivots so that the claw portion **401** rises when the energy storage device **40** moves by self-propelling, so that the claw portion **401** is located above a lower end of the wheel **42** of the energy storage device **40**. Then, when the energy storage device **40** is stopped inside the accommodation portion **31** of the moving body **30**, the claw portion **401** pivots so as to be lowered and the claw portion **401** is arranged in the slit groove **311** of the accommodation portion **31** of the moving body **30**. Further, the claw portion **401** moves away from the side surface to which the fixing member **400** of the energy storage device **40** is attached and is engaged with the slit groove **311**. Then, the energy storage device **40** is fixed to the floor **310** of the accommodation portion **31** of the moving body **30** by engaging the claw portion **401** of the fixing member **400** with the slit groove **311**. Then, when the claw portion **401** moves so as to approach the side surface to which the fixing member **400** of the energy storage device **40** is attached, the claw portion **401** is disengaged from the slit groove **311**. In this way, the claw portion **401** of the fixing member **400** of the energy storage device **40** can be engaged with and disengaged from the slit groove **311** of the accommodation portion **31** of the moving body **30**.

(109) As a result, the energy storage device **40** can be quickly engaged and disengaged and firmly fixed to the accommodation portion **31** of the moving body **30** without human intervention. Further, when the energy storage device **40** moves by self-propelling, the claw portion **401** pivots so as to rise and the claw portion **401** is located above the lower end of the wheel **42** of the energy storage device **40**. Therefore, the fixing member **400** can be prevented from interfering with the movement of the energy storage device **40**.

(110) A width of the slit groove **311** is sufficiently narrower than the wheel **42** of the energy storage device **40** so that the wheel **42** of the energy storage device **40** does not fall into the slit groove **311**.

(111) In the embodiment, an engagement sensor **311A** is provided in the slit groove **311** of the accommodation portion **31** of the moving body **30**. The engagement sensor **311A** is a sensor which detects whether the energy storage device **40** is engaged with the slit groove **311** and is, for example, a pressure sensor. Further, the engagement sensor **311A** transmits information on whether the energy storage device **40** is engaged with the slit groove **311** to the third information management terminal of the moving body **30**. As a result, the moving body **30** and the central information management server **60** can acquire information on whether the energy storage device **40** is engaged with the slit groove **311**.

(112) In the embodiment, the engagement sensor **311A** is provided in the slit groove **311** of the accommodation portion **31** of the moving body **30**, but it may be provided in the fixing member **400** of the energy storage device **40**. Also, the engagement sensor **311A** may be provided in both the slit groove **311** of the accommodation portion **31** of the moving body **30** and the fixing member **400** of the energy storage device **40**.

(113) As illustrated in FIGS. **6A** to **6C**, disengageable fixation of the energy storage device **40** to the accommodation portion **31** of the moving body **30** may be performed by a lock pin **402** provided in the energy storage device **40** and a lock hole **312** provided in the floor **310** of the accommodation portion **31** of the moving body **30**.

(114) As illustrated in FIG. **6A**, the energy storage device **40** includes the lock pin **402** which projects toward the floor **310** of the accommodation portion **31** of the moving body **30**. The floor **310** of the accommodation portion **31** of the moving body **30** is provided with the lock hole **312** into which the lock pin **402** can be inserted.

(115) As illustrated in FIG. **6B**, the lock pin **402** can be moved up and down with respect to the energy storage device **40**. When the energy storage device **40** moves by self-propelling, the lock pin **402** is raised so as to be located above the lower end of the wheel **42** of the energy storage device **40**. Then, when the energy storage device **40** is stopped inside the accommodation portion **31** of the moving body **30**, the lock pin **402** is lowered and inserted into the lock hole **312** of the accommodation portion **31** of the moving body **30**, and then is engaged with the lock hole **312**. Then, the energy storage device **40** is fixed to the floor **310** of the accommodation portion **31** of the moving body **30** by engaging the lock pin **402** with the lock hole **312**. Then, when the lock pin **402** rises, the engagement with the lock hole **312** is disengaged. In this way, the lock pin **402** of the energy storage device **40** can be engaged with and disengaged from the lock hole **312** of the accommodation portion **31** of the moving body **30**.

(116) As illustrated in FIG. **6C**, the entire energy storage device **40** may be movable up and down except for the wheel **42**. When the energy storage device **40** moves by self-propelling, the energy storage device **40** is raised as a whole except for the wheel **42** and the lock pin **402** is located above the lower end of the wheel **42** of the energy storage device **40**. Then, when the energy storage device **40** is stopped inside the accommodation portion **31** of the moving body **30**, the entire energy storage device **40** is lowered except for the wheel **42** and the lock pin **402** is inserted into the lock hole **312** of the accommodation portion **31** of the moving body **30** and is engaged with the lock hole **312**. Then, the energy storage device **40** is fixed to the floor **310** of the accommodation portion **31** of the moving body **30** by engaging the lock pin **402** with the lock hole **312**. Then, when the energy storage device **40** rises as a whole except for the wheel **42**, the engagement with the lock hole **312** is disengaged. In this way, the lock pin **402** of the energy storage device **40** may be disengageable from the lock hole **312** of the accommodation portion **31** of the moving body **30**.

(117) As a result, the energy storage device **40** can be quickly engaged and disengaged and firmly fixed to the accommodation portion **31** of the moving body **30** without human intervention. In addition, when the energy storage device **40** moves by self-propelling, the lock pin **402** is located above the lower end of the wheel of the energy storage device **40**. Therefore, the lock pin **402** can

be prevented from interfering with the movement of the energy storage device **40**.

(118) In addition, an engagement sensor **312A** is provided in the lock hole **312** of the accommodation portion **31** of the moving body **30**. The engagement sensor **312A** is a sensor which detects whether the energy storage device **40** is inserted into the lock hole **312** and is, for example, a pressure sensor. In addition, the engagement sensor **312A** transmits information on whether the energy storage device **40** is engaged with the lock hole **312** to the third information management terminal of the moving body **30**. As a result, the moving body **30** and the central information management server **60** can acquire information on whether the energy storage device **40** is engaged with the lock hole **312**.

(119) In the embodiment, it is assumed that the engagement sensor **312A** is provided in the lock hole **312** of the accommodation portion **31** of the moving body **30**, but may be provided in the lock pin **402** of the energy storage device **40**. Also, the engagement sensor **312A** may be provided in both the lock hole **312** of the accommodation portion **31** of the moving body **30** and the lock pin **402** of the energy storage device **40**.

Second Embodiment

(120) Subsequently, an energy storage device transport system **1A** according to a second embodiment of the invention will be described with reference to FIG. 7. In the following description, the same components as those of the energy storage device transport system **1** of the first embodiment are designated by the same numerals and letters and the description thereof is omitted or simplified. In the energy storage device transport system **1** of the first embodiment, it is assumed that the energy storage device **40** is loaded with the transported object **70** such as luggage. However, in the energy storage device transport system **1A** of the second embodiment, the energy storage device **40** is not loaded with the transported object **70**. In the energy storage device transport system **1A** of the second embodiment, the energy stored in the energy storage unit **41** of the energy storage device **40** is transported to the second station **20**, which is the transport destination. That is, the transported object transported by the energy storage device transport system **1** of the first embodiment to the second station **20** is the transported object **70** such as luggage and the transported object transported by the energy storage device transport system **1A** of the second embodiment to the second station **20** is energy. In the energy storage device transport system **1** of the first embodiment, the transported object **70** transported to the second station **20** may be an energy source. Hereinafter, differences between the energy storage device transport system **1** of the first embodiment and the energy storage device transport system **1A** of the second embodiment will be described in detail.

(121) The energy storage device transport system **1A** of the second embodiment can transport the energy to the second station **20** by transporting the energy storage device **40** in which the energy is stored in the energy storage unit **41** to the second station **20**.

(122) The energy storage device **40** is equipped with a connection terminal such as an outlet plug of the related art. Therefore, by connecting an electronic device or the like to the connection terminal, the energy storage device **40** can operate the electronic device or the like by using the energy stored in the energy storage unit **41** of the energy storage device **40**. Then, the energy stored in the energy storage unit **41** of the energy storage device **40** is used around the second station **20**.

(123) The energy stored in the energy storage unit **41** of the energy storage device **40** around the second station **20**, which is the transport destination of the energy storage device **40**, is expected to be used in various ways. That is, the energy storage unit **41** of the energy storage device **40** is required to have various output characteristics. In response to such demands, it is difficult to increase the C rate in Li-ion batteries of the related art because the diffusion rate of solvated ions is limited. On the other hand, it is possible to handle even higher C rates because all-solid-state batteries are less susceptible to the restrictions of solvated ion diffusion. In addition, since the all-solid-state battery does not have an electrolyte, the growth of the SEI layer, which is considered to be derived from the electrolytic solution and is the main cause of the deterioration of the Li-ion

battery, is suppressed, so that the deterioration is small.

(124) Further, the energy storage unit **41** of the energy storage device **40** of the embodiment may be used repeatedly or may be stored at a high potential for a long time. Therefore, using an all-solid-state battery for the energy storage unit **41** of the energy storage device **40** reduces the frequency of replacement due to deterioration, which is advantageous in terms of cost.

(125) Therefore, the energy storage unit **41** of the energy storage device **40** in the energy storage device transport system **1A** of the embodiment is preferably an all-solid-state battery.

(126) The second energy management unit **22** of the second station **20** transmits an energy demand amount around the second station **20** to the central information management server **60**. The central information management server **60** transmits a remaining required energy amount of the energy storage unit **41** of the energy storage device **40** to be transported, to the second station **20** to the third energy management unit **32** of the moving body **30** based on the energy demand amount around the second station **20**. The energy demand amount transmitted by the second energy management unit **22** of the second station **20** to the central information management server **60** may be the predicted energy demand amount calculated by using, for example, a mechanical learning method, an artificial intelligence method, or the like.

(127) Then, based on the remaining required energy amount of the energy storage unit **41** of the energy storage device **40** to be transported to the second station **20**, which is transmitted to the third energy management unit **32**, and the remaining energy state (SOC) of the energy storage unit **41** of each energy storage device **40**, which is acquired by the energy storage unit state acquisition units **444** of a plurality of energy storage devices **40** accommodated in the accommodation portion **31** of the moving body **30**, each energy storage device **40** changes a position inside the accommodation portion **31** of the moving body **30** by self-propelling.

(128) In the embodiment, based on the remaining required energy amount of the energy storage unit **41** of the energy storage device **40** to be transported to the second station **20**, the energy storage device **40** with the optimal remaining energy state (SOC) of the energy storage unit **41** moves to the vicinity of the slope **30S** of the accommodation portion **31** by self-propelling.

(129) Therefore, for example, when the energy demand amount around the second station **20** is high and the moving body **30** approaches the second station **20**, the energy storage device **40** in which the remaining energy state (SOC) of the energy storage unit **41** is fully charged can be moved to a position where it is easy to move to the outside of the accommodation portion **31** of the moving body **30**.

(130) In this way, when the moving body **30** approaches the second station **20**, the optimal energy storage device **40** moves to a position where it is easy to move to the outside of the accommodation portion **31** of the moving body **30**. Therefore, inside the accommodation portion **31** of the moving body **30**, the energy storage device **40** can be optimally arranged according to the energy demand amount at the second station **20** and the remaining amount of energy stored in the energy storage unit **41**.

(131) A self-propelling control command to the energy storage device **40** inside the accommodation portion **31** of the moving body **30** may be a control command generated by the third energy management unit **32**, or may be a control command from the central information management server **60**.

(132) Although the embodiments of the invention are described above, the invention is not limited to the embodiments described above and can be appropriately modified, improved, and the like.

(133) For example, the energy storage device **40** may perform the life diagnosis of the energy storage unit **41** by acquiring the deterioration information and the usage history of the energy storage unit **41** with the energy storage unit state acquisition unit **444**. Then, when the energy storage unit **41** needs to be replaced, the energy storage device **40** may transmit information to the central information management server **60** that the energy storage unit **41** needs to be replaced and the central information management server **60** may control the movement of the moving body **30**

and each energy storage device **40** so as to transport the energy storage device **40** to a recycling center. This will reduce the cost and man-hours required to replace the energy storage unit **41**.

(134) Further, for example, the central information management server **60** may be managed by blockchain technology. In addition, the central information management server **60** may optimize the overall energy and/or logistics system of the energy storage device transport system **1** and **1A** each of which includes the first station **10**, the second station **20**, the moving body **30**, each energy storage device **40**, and the system power grid **50** using mechanical learning methods and artificial intelligence methods and may control the movement of the moving body **30** and each energy storage device **40**.

(135) Further, in the first embodiment and the second embodiment, it is assumed that the moving body **30** is moved by unmanned operation in which the drive unit **33** is driven by the electric power output from the control output unit **367** of the third information management terminal **36**, but the moving body **30** may be moved by manned operation.

(136) Further, in the first embodiment and the second embodiment, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** is self-propelled inside the accommodation portion **31** of the moving body **30** based on the information of the traveling state which is at least one of the velocity and the acceleration of the moving body **30** acquired by the traveling state acquisition unit **363** of the moving body **30**. However, the energy storage device **40** may acquire the traveling state of the moving body **30** based on an operating state such as whether the moving body **30** is turned on, not limited to at least one of the velocity and the acceleration of the moving body **30** acquired by the traveling state acquisition unit **363** of the moving body **30** and may be self-propelled inside the accommodation portion **31** of the moving body **30**.

(137) Further, in the first embodiment and the second embodiment, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** acquires the traveling state of the moving body **30** from the information of at least one of the velocity and the acceleration of the moving body **30** received by the wireless communication unit **45** and is self-propelled inside the accommodation portion **31** of the moving body **30** based on the acquired traveling state of the moving body **30**. However, the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** may acquire the traveling state of the moving body **30** from the information of at least one of the velocity and the acceleration detected by the traveling state acquisition unit **443** and may be self-propelled inside the accommodation portion **31** of the moving body **30** based on the acquired traveling state of the moving body **30**. In this case, the energy storage device **40** can be self-propelled inside the accommodation portion **31** of the moving body **30** based on the information of at least one of the estimated velocity and acceleration of the moving body **30** without acquiring the information of at least one of the velocity and the acceleration of the moving body **30** from the moving body **30**.

(138) Further, in the first embodiment and the second embodiment, the information of at least one of the velocity and the acceleration of the moving body **30** acquired by the traveling state acquisition unit **363** of the moving body **30** is transmitted from the wireless communication unit **368** of the third information management terminal **36** and the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** acquires the traveling state of the moving body **30**. However, the moving body **30** may acquire the traveling state of the moving body **30** and generate a control signal of the energy storage device **40** accommodated in the accommodation portion **31** based on the acquired traveling state of the moving body **30**. Then, the control signal of the energy storage device **40** may be transmitted from the wireless communication unit **368** of the moving body **30** and the energy storage device **40** accommodated in the accommodation portion **31** of the moving body **30** may receive the control signal of the energy storage device **40** with the wireless communication unit **45** and be self-propelled inside the accommodation portion **31** of the moving body **30** by the received control signal of the energy storage device **40**.

(139) Further, the energy storage device **40** may have a plurality of energy storage units **41**. Some energy storage units **41** may be fixed to the energy storage device **40**. In addition, some energy storage units **41** are detachably provided in the energy storage device **40** and may be loaded and unloaded with respect to the energy storage device **40**.

(140) Then, at least one of the transport source and transport destination of the energy storage device **40**, the energy storage device **40** may unload the energy storage unit **41** from the energy storage device **40** or load the energy storage unit **41** on the energy storage device **40**.

(141) In addition, the transportation of the energy storage device **40** described in the first embodiment and the second embodiment may be executed by a moving body transport program including an acquisition step of acquiring at least one of the moving state and the operating state of the moving body **30** recorded on a recording medium and a repositioning step of repositioning the energy storage device **40** in the accommodation portion **31** by making the energy storage device **40** self-propel inside the accommodation portion **31** based on at least one of the moving state and the operating state of the moving body **30** acquired in the acquisition step.

(142) In addition, at least the following matters are described in this specification. The components and the like corresponding to those of the embodiments described above are shown in parentheses, but the invention is not limited to these.

(143) (A1) A transport system (energy storage device transport system **1**, **1A**) which includes a first moving body (energy storage device **40**) which can move autonomously and a second moving body (moving body **30**) which has an accommodation portion (accommodation portion **31**) for accommodating the first moving body and in which the second moving body accommodates the first moving body and moves to transport the first moving body, where the first moving body autonomously moves inside the accommodation portion of the second moving body and changes a position in the accommodation portion of the second moving body.

(144) According to (A1), the first moving body autonomously moves inside the accommodation portion of the second moving body and changes the position in the accommodation portion of the second moving body. Therefore, the first moving body can be quickly arranged and fixed in the optimal position in the accommodation portion of the second moving body without human intervention.

(145) (A2) The transport system according to (A1), where the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on at least one of a moving state and an operating state of the second moving body.

(146) According to (A2), the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on the moving state and the operating state of the second moving body. Therefore, the first moving body can change the position by autonomously moving inside the accommodation portion of the second moving body in a safe state.

(147) (A3) The transport system according to (A2), where the moving state of the second moving body is at least one of velocity and acceleration of the second moving body.

(148) According to (A3), since the moving state of the second moving body is at least one of the velocity and the acceleration of the second moving body, the first moving body can change the position by autonomously moving inside the accommodation portion of the second moving body based on at least one of the velocity and the acceleration of the second moving body. As a result, the first moving body can change the position by autonomously moving inside the accommodation portion of the second moving body in a safe state with a simple configuration.

(149) (A4) The transport system according to (A3), where the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body when at least one of the velocity and the acceleration of the second moving body is equal to or less than a predetermined value.

(150) According to (A4), the first moving body changes the position by autonomously moving

inside the accommodation portion of the second moving body when at least one of the velocity and the acceleration of the second moving body is equal to or less than the predetermined value.

Therefore, the first moving body can change the position by autonomously moving inside the accommodation portion of the second moving body in a safer state.

(151) (A5) The transport system according to any one of (A1) to (A4), where the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on a movement environment around the second moving body.

(152) According to (A5), the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on the movement environment around the second moving body. Therefore, the first moving body can change the position by autonomously moving inside the accommodation portion of the second moving body in a safe state.

(153) (A6) The transport system according to (A5), where the movement environment around the second moving body is at least one of traffic information around the second moving body and moving infrastructure information around the second moving body.

(154) According to (A6), the movement environment around the second moving body is at least one of the traffic information around the second moving body and the moving infrastructure information around the second moving body. Therefore, the first moving body can change the position by autonomously moving inside the accommodation portion of the second moving body based on at least one of the traffic information around the second moving body and the moving infrastructure information around the second moving body. As a result, the first moving body can change the position by autonomously moving inside the accommodation portion of the second moving body in a safe state with a simple configuration.

(155) (A7) The transport system according to any one of (A1) to (A6), where the first moving body, includes a first energy storage unit (energy storage unit 41) and transports energy stored in the first energy storage unit from a transport source to a transport destination, and at least one of the transport source and the transport destination, the first moving body, outputs the energy stored in the first energy storage unit to the outside, inputs external energy to the first energy storage unit, unloads the first energy storage unit from the first moving body, or loads the first energy storage unit on the first moving body.

(156) According to (A7), the first moving body includes the first energy storage unit and can transport the energy stored in the first energy storage unit from the transport source to the transport destination. Further, at least one of the transport source and the transport destination, the first moving body can output the energy stored in the first energy storage unit to the outside, input external energy to the first energy storage unit, unload the first energy storage unit from the first moving body, or load the first energy storage unit on the first moving body.

(157) (A8) The transport system according to (A7), where the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on a remaining amount of energy stored in the first energy storage unit.

(158) According to (A8), the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on the remaining amount of energy stored in the first energy storage unit. Therefore, inside the accommodation portion of the second moving body, the first moving body can be optimally arranged according to the remaining amount of energy stored in the first energy storage unit.

(159) (A9) The transport system according to (A7) or (A8), where the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on an energy demand amount of the transport destination.

(160) According to (A9), the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on the energy demand amount of the transport destination. Therefore, inside the accommodation portion of the second moving body, the first moving body can be optimally arranged according to the energy demand amount of

the transport destination.

(161) (A10) The transport system according to any one of (A7) to (A9), where the second moving body includes a second energy storage unit (energy storage unit 34), and the first energy storage unit of the first moving body and the second energy storage unit of the second moving body are provided so that energy can be transmitted and received to and from each other.

(162) According to (A10), the first energy storage unit of the first moving body and the second energy storage unit of the second moving body are provided so that energy can be transmitted and received to and from each other. Therefore, the energy stored in the first energy storage unit of the first moving body and the energy stored in the second energy storage unit of the second moving body can be efficiently used.

(163) (A11) The transport system according to (A10), where the accommodation portion of the second moving body has an energy transmission and reception area (energy transmission and reception area 31A) where the first energy storage unit and the second energy storage unit transmit and receive energy to and from each other, and the first moving body autonomously moves inside the accommodation portion of the second moving body and changes the position so that the position is within the energy transmission and reception area when the first energy storage unit and the second energy storage unit transmit and receive energy to and from each other.

(164) According to (A11), the first moving body autonomously moves inside the accommodation portion of the second moving body and changes the position so that the position is within the energy transmission and reception area when the first energy storage unit and the second energy storage unit transmit and receive energy to and from each other. Therefore, even when the transport system includes a plurality of first moving bodies, energy can be efficiently transferred between the first energy storage unit of the first moving body and the second energy storage unit of the second moving body without human intervention.

(165) (A12) The transport system according to any one of (A7) to (A11), where the accommodation portion of the second moving body is provided so as to be able to accommodate a plurality of the first moving bodies, and the first energy storage unit of the first moving body is provided so as to be able to transmit and receive energy to and from the first energy storage unit of a different first moving body.

(166) According to (A12), the accommodation portion of the second moving body is provided so as to be able to accommodate a plurality of the first moving bodies and the first energy storage unit of the first moving body is provided so as to be able to transmit and receive energy to and from the first energy storage unit of a different first moving body. Therefore, the energy stored in the first energy storage units of the plurality of first moving bodies accommodated in the accommodation portion of the second moving body can be stored in the first energy storage units of each of the first moving bodies with the optimal distribution.

(167) (A13) The transport system according to any one of (A1) to (A12), where the first moving body includes a loading unit (loading unit 46) which loads a transported object (transported object 70).

(168) According to (A13), the first moving body includes the loading unit which loads the transported object. Therefore, the first moving body can load the transported object such as luggage on the loading unit and transport the transported object from the transport source to the transport destination.

(169) (A14) The transport system according to (A13), where the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on transport destination information of the transported object.

(170) According to (A14), the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on the transport destination information of the transported object. Therefore, when the moving body approaches the transport destination of the transported object, the first moving body can be moved to a position

where it is easy to move to the outside of the accommodation portion of the second moving body and the transported object can be efficiently moved from the accommodation portion of the second moving body.

(171) (A15) The transport system according to (A13) or (A14), where the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on content information of the transported object.

(172) According to (A15), the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on the content information of the transported object. Therefore, for example, when the transported object article loaded on the loading unit is a fragile item, the movement of the first moving body inside the accommodation portion of the second moving body can be adjusted according to the contents of the transported object loaded on the loading unit, such as not moving autonomously and changing the position in the accommodation portion of the second moving body.

(173) (A16) The transport system according to any one of (A1) to (A15), where the second moving body includes an auxiliary portion (slope 30S) which allows the first moving body to be able to autonomously move between the inside and the outside of the accommodation portion.

(174) According to (A16), the second moving body includes the auxiliary portion which allows the first moving body to be able to autonomously move between the inside and the outside of the accommodation portion. Therefore, the first moving body can be moved from the outside of the second moving body to the inside of the accommodation portion of the second moving body by autonomous movement. Also, the first moving body can be moved from the inside of the accommodation portion of the second moving body to the transport destination by autonomous movement. Therefore, the work of loading the first moving body into the second moving body and unloading the first moving body from the second moving body to the transport destination after the second moving body arrives around the transport destination, which is performed manually in the related art, can be performed without human intervention.

(175) (A17) The transport system according to any one of (A1) to (A16), where the first moving body has a first engagement portion (claw portion 401, lock pin 402), the accommodation portion of the second moving body has a second engagement portion (slit groove 311, lock hole 312), and the first engagement portion and the second engagement portion are engaged with each other in a disengageable manner.

(176) According to (A17), the first engagement portion of the first moving body and the second engagement portion of the accommodation portion are engaged with each other in a disengageable manner. Therefore, the first moving body can be fixed to the accommodation portion.

(177) (A18) The transport system according to (A17), where the first engagement portion is provided so as to be displaceable between an engaged position and a non-engaged position, and is engaged with the second engagement portion at the engaged position in a state where the first moving body is stopped.

(178) According to (A18), at the time of autonomous movement of the first moving body, the first engagement portion is located in the non-engaged position, and in the state where the first moving body is stopped, the first engagement portion is located in the engaged position and is engaged with the second engagement portion. Therefore, at the time of the autonomous movement of the first moving body, the first engagement portion can be prevented from interfering with the autonomous movement of the first moving body.

(179) (A19) The transport system according to (A17) or (A18), where the second engagement portion has a slit groove (slit groove 311) recessed substantially linearly in a floor of the accommodation portion, and the first engagement portion has a claw portion (claw portion 401) which is engaged with the slit groove.

(180) According to (A19), the second engagement portion has the slit groove recessed substantially linearly in the floor of the accommodation portion and the first engagement portion has the claw

portion which is engaged with the slit groove. Therefore, the first moving body can be quickly engaged and disengaged and firmly fixed to the floor of the accommodation portion.

(181) (A20) The transport system according to any one of (A1) to (A19), where the first moving body, is a vehicle having a wheel, and can autonomously travel inside the accommodation portion of the second moving body.

(182) According to (A20), the first moving body is a vehicle having a wheel and can autonomously travel inside the accommodation portion of the second moving body. Therefore, the first moving body can be easily arranged and fixed in the optimal position of the accommodation portion of the second moving body without human intervention.

(183) (A21) A moving body (energy storage device 40) capable of autonomously moving which is accommodated in an accommodation portion (accommodation portion 31) of another moving body (moving body 30) and transported by the other moving body, including: an acquiring unit which acquires at least one of a moving state and an operating state of the other moving body; and a position changing unit which autonomously moves inside the accommodation portion and changes a position in the accommodation portion based on at least one of the moving state and the operating state of the other moving body acquired by the acquiring unit.

(184) According to (A21), the moving body autonomously moves inside the accommodation portion of the other moving body and changes the position in the accommodation portion of the other moving body. Therefore, the moving body can be quickly arranged and fixed in the optimal position of the accommodation portion of the other moving body without human intervention.

(185) (A22) A transport method in which an autonomously movable moving body (energy storage device 40) is accommodated in an accommodation portion (accommodation portion 31) of another moving body (moving body 30) and transported by the other moving body, including the steps of: acquiring at least one of a moving state and an operating state of the other moving body; and changing a position of the moving body in the accommodation portion by autonomously moving the moving body inside the accommodation portion based on at least one of the moving state and the operating state of the other moving body acquired in the step of acquiring.

(186) According to (A22), the position of the moving body in the accommodation portion of the other moving body can be changed by autonomously moving the moving body inside the accommodation portion of the other moving body. Therefore, the moving body can be quickly arranged and fixed in the optimal position of the accommodation portion of the other moving body without human intervention.

(187) (A23) A transport program for accommodating an autonomously movable moving body (energy storage device 40) in an accommodation portion (accommodation portion 31) of another moving body (moving body 30) and transporting the moving body by the other moving body, including the steps of; acquiring at least one of a moving state and an operating state of the other moving body; and changing a position of the moving body in the accommodation portion by autonomously moving the moving body inside the accommodation portion based on at least one of the moving state and the operating state of the other moving body acquired in the step of acquiring.

(188) According to (A23), the position of the moving body in the accommodation portion of the other moving body can be changed by autonomously moving the moving body inside the accommodation portion of the other moving body. Therefore, the moving body can be quickly arranged and fixed in the optimal position of the accommodation portion of the other moving body without human intervention.

(189) (A24) A recording medium on which the transport program according to (A23) is recorded.

(190) According to (A24), the transport program can be easily executed by causing a computer to read the recording medium on which the transport program is recorded.

(191) This application is based on a Japanese patent application filed on Mar. 29, 2019 (Japanese Patent Application No. 2019-068616), the contents of which are incorporated herein by reference.

DESCRIPTION OF REFERENCE NUMERALS AND CHARACTERS

(192) **1**, **1A**: energy storage device transport system (transport system) **30**: moving body (second moving body, another moving bodies) **30S**: slope (auxiliary portion) **31**: accommodation portion **31A**: energy transmission and reception area **311**: slit groove (second engagement portion) **312**: lock hole (second engagement portion) **34**: energy storage unit (second energy storage unit) **40**: energy storage device (first moving body, moving body) **401**: claw portion (first engagement portion) **402**: lock pin (first engagement portion) **41**: energy storage unit (first energy storage unit) **46**: loading unit **70**: transported object

Claims

1. A transport system comprising: a first moving body which can move autonomously; and a second moving body which has an accommodation portion for accommodating the first moving body, wherein: the second moving body accommodates the first moving body and moves to transport the first moving body; the first moving body autonomously moves from a first position inside the accommodation portion of the second moving body to a second position as a destination inside the accommodation portion and changes a position inside the accommodation portion of the second moving body; the first moving body includes a first wireless communication unit; the second moving body includes a traveling state acquisition unit configured to acquire information on a traveling state of the second moving body, and a second wireless communication unit configured to transmit the information on the traveling state acquired by the traveling state acquisition unit; the information on the traveling state of the second moving body is at least one of velocity and acceleration of the second moving body; and the first moving body changes the position during the movement of the second moving body, when at least one of the velocity and the acceleration of the second moving body is equal to or less than a predetermined value, by: autonomously moving inside the accommodation portion of the second moving body based on the traveling state of the second moving body indicated in the information received by the first wireless communication unit; or autonomously moving inside the accommodation portion of the second moving body according to a control signal received by the first wireless communication unit when the first wireless communication unit receives the control signal generated based on the traveling state of the second moving body.
2. The transport system according to claim 1, wherein the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on a movement environment around the second moving body.
3. The transport system according to claim 2, wherein the movement environment around the second moving body is at least one of traffic information around the second moving body and moving infrastructure information around the second moving body.
4. The transport system according to claim 1, wherein: the first moving body includes a first energy storage unit and transports energy stored in the first energy storage unit from a transport source to a transport destination; and at least one of the transport source and the transport destination, the first moving body, outputs the energy stored in the first energy storage unit to the outside, inputs external energy to the first energy storage unit, unloads the first energy storage unit from the first moving body, or loads the first energy storage unit on the first moving body.
5. The transport system according to claim 4, wherein the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on a remaining amount of energy stored in the first energy storage unit.
6. The transport system according to claim 4, wherein the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on an energy demand amount of the transport destination.
7. The transport system according to claim 4, wherein: the second moving body includes a second energy storage unit; and the first energy storage unit of the first moving body and the second

energy storage unit of the second moving body are provided so that energy can be transmitted and received to and from each other.

8. The transport system according to claim 7, wherein: the accommodation portion of the second moving body has an energy transmission and reception area where the first energy storage unit and the second energy storage unit transmit and receive energy to and from each other; and the first moving body autonomously moves inside the accommodation portion of the second moving body and changes the position so that the position is within the energy transmission and reception area when the first energy storage unit and the second energy storage unit transmit and receive energy to and from each other.

9. The transport system according to claim 4, wherein: the accommodation portion of the second moving body is provided so as to be able to accommodate a plurality of the first moving bodies; and the first energy storage unit of the first moving body is provided so as to be able to transmit and receive energy to and from the first energy storage unit of a different first moving body.

10. The transport system according to claim 1, wherein the first moving body includes a loading unit which loads a transported object.

11. The transport system according to claim 10, wherein the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on transport destination information of the transported object.

12. The transport system according to claim 10, wherein the first moving body changes the position by autonomously moving inside the accommodation portion of the second moving body based on content information of the transported object.

13. The transport system according to claim 1, wherein the second moving body includes an auxiliary portion which allows the first moving body to be able to autonomously move between the inside and the outside of the accommodation portion.

14. The transport system according to claim 1, wherein: the first moving body has a first engagement portion; the accommodation portion of the second moving body has a second engagement portion; and the first engagement portion and the second engagement portion are engaged with each other in a disengageable manner.

15. The transport system according to claim 14, wherein the first engagement portion is provided so as to be displaceable between an engaged position and a non-engaged position, and is engaged with the second engagement portion at the engaged position in a state where the first moving body is stopped.

16. The transport system according to claim 14, wherein: the second engagement portion has a slit groove recessed substantially linearly in a floor of the accommodation portion; and the first engagement portion has a claw portion which is engaged with the slit groove.

17. The transport system according to claim 1, wherein: the first moving body is a vehicle having a wheel, and can autonomously travel inside the accommodation portion of the second moving body.

18. The transport system according to claim 1, wherein the first moving body starts autonomously moving from the first position in a state where the first moving body is stopped, and stops at the second position as the destination.

19. A moving body capable of autonomously moving which is accommodated in an accommodation portion of another moving body and transported by the other moving body, comprising: a position changing unit which autonomously moves from a first position inside the accommodation portion of the other moving body to a second position as a destination inside the accommodation portion of the other moving body and changes a position inside the accommodation portion of the other moving body, wherein: the moving body includes a first wireless communication unit; the other moving body includes a traveling state acquisition unit configured to acquire information on a traveling state of the other moving body, and a second wireless communication unit configured to transmit the information on the traveling state acquired by the traveling state acquisition unit or transmit a control signal for the moving body, the control

signal being generated based on information on the traveling state acquired by the traveling state acquisition unit; the information on the traveling state acquired by the traveling state acquisition unit is at least one of velocity and acceleration of the other moving body; and the moving body changes the position during the movement of the other moving body, when at least one of the velocity and the acceleration of the other moving body is equal to or less than a predetermined value, by: autonomously moving inside the accommodation portion of the other moving body based on the traveling state of the other moving body indicated in the information received by the first wireless communication unit; or autonomously moving inside the accommodation portion of the other moving body according to the control signal received by the first wireless communication unit when the first wireless communication unit receives the control signal generated based on the traveling state of the other moving body.

20. A transport method in which an autonomously movable moving body is accommodated in an accommodation portion of another moving body and transported by the other moving body, comprising: a position changing step for changing a position of the moving body inside the accommodation portion of the other moving body by autonomously moving the moving body from a first position inside the accommodation portion of the other moving body to a second position as a destination inside the accommodation portion of the other moving body; an acquiring step for acquiring, by a traveling state acquisition unit of the other moving body, information on a traveling state of the other moving body; and a transmitting step for transmitting, by a second wireless communication unit of the other moving body, the information on the traveling state of the other moving body or transmitting a control signal for the moving body, the control signal being generated based on the information on the traveling state of the other moving body, wherein the information on the traveling state of the other moving body is at least one of velocity and acceleration of the other moving body, and wherein, in the position changing step, the moving body changes the position during the movement of the other moving body, when at least one of the velocity and the acceleration of the other moving body is equal to or less than a predetermined value, by: autonomously moving inside the accommodation portion of the other moving body based on the traveling state of the other moving body indicated in the information acquired in the acquiring step and received by a first wireless communication unit of the moving body; or autonomously moving inside the accommodation portion of the other moving body according to the control signal received by a first wireless communication unit of the moving body when the first wireless communication unit receives the control signal generated based on the traveling state of the other moving body.

21. A computer program product, for accommodating an autonomously movable moving body in an accommodation portion of another moving body and transporting the moving body by the other moving body, comprising a non-transitory computer readable storage medium having instructions encoded thereon that, when executed by a processor, cause the processor to execute a process, the process comprising: a position changing step for changing a position of the moving body inside the accommodation portion of the other moving body by autonomously moving the moving body from a first position inside the accommodation portion of the other moving body to a second position as a destination inside the accommodation portion of the other moving body; an acquiring step for acquiring, by a traveling state acquisition unit of the other moving body, information on a traveling state of the other moving body; and a transmitting step for transmitting, by a second wireless communication unit of the other moving body, the information on the traveling state of the other moving body or transmitting a control signal for the moving body, the control signal being generated based on the information on the traveling state of the other moving body, wherein the information on the traveling state of the other moving body is at least one of velocity and acceleration of the other moving body, and wherein, in the position changing step, the moving body changes the position during the movement of the other moving body, when at least one of the velocity and the acceleration of the other moving body is equal to or less than a predetermined

value, by: autonomously moving inside the accommodation portion of the other moving body based on the traveling state of the other moving body indicated in the information acquired in the acquiring step and received by a first wireless communication unit of the moving body, or autonomously moving inside the accommodation portion of the other moving body according to the control signal received by a first wireless communication unit of the moving body when the first wireless communication unit receives the control signal generated based on the traveling state of the other moving body.
